

Backtesting Probability of Default Models

What is the Statistical Hypothesis Behind Z-Score, Exact Binomial and Jeffreys Tests?

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Probability of Default Backtesting

- As risk quantification is a regulatory requirement, practitioners pay significant attention to complying with these requirements.
- In addition to following the prescribed risk quantification process, practitioners must assess the appropriateness of model estimates. This is typically referred to as backtesting.
- The primary purpose of backtesting in this context is to ensure that model estimates accurately predict the occurrence of defaults, i.e., that Probability of Default (PD) estimates provide reliable forecasts of default rates.
- In practice, backtesting is one of the most comprehensive evaluation and validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Backtesting often includes a set of quantitative and qualitative tools used to draw an overall conclusion on the appropriateness of PD model estimates.
- One important aspect of the quantitative tools is the statistical testing framework commonly used for backtesting.

Statistical Tests

- As part of the quantitative testing framework for backtesting, practitioners typically use the following three tests:
 - exact binomial test;
 - z-score test;
 - Jeffreys test.
- Given the conservative nature of IRB modelling, hypothesis tests are typically formulated to assess whether there is evidence of possible underestimation.
- In simple terms, testing for underestimation means that practitioners employ one-sided versions of the tests. Nevertheless, certain evaluation and validation aspects also require testing for accuracy or precision, which involves two-sided testing.
- The above-mentioned tests are typically used as alternatives for underestimation testing. However, practitioners often overlook that these tests rely on different underlying testing assumptions.
- The purpose of this presentation is to demonstrate, through visualisation, the testing hypotheses behind each of these three tests. The simulation results are based on testing for evidence of underestimation, but the points regarding differences in the underlying testing hypotheses remain valid for the two-sided versions as well.

Exact binomial

Given a confidence level q , for underestimation testing, the null hypothesis is rejected if the number of defaults k in the tested segment is greater than or equal to a critical value k^* , defined as:

$$k^* = \min \left\{ k \mid \sum_{nd=k}^{no} \binom{no}{nd} PD^{nd} (1 - PD)^{no - nd} \leq 1 - q \right\}$$

where:

- no is the number of obligors;
- PD is the calibrated PD.

Using the binomial distribution function, practitioners can express the results of the exact binomial test in terms of a p-value based on the actual number of observed defaults. This approach is commonly used in practice, as it provides results that are directly comparable with those of other statistical tests.

$$Pr(X \geq d) = \binom{no}{nd} PD^{nd} (1 - PD)^{no - nd}$$

where nd is the number of defaults.

It should be noted that the exact binomial test follows the frequentist approach, assuming that the calibrated PD represents the true PD.

Z-Score Test

The test statistic for the z-score test is given as follows:

$$Z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{no}}}$$

where:

- *ODR* is the observed default rate;
- *PD* is the calibrated PD;
- *no* is the number of observations.

Under the assumption that the Z_{score} test statistic follows the standard normal distribution, a p-value is calculated accordingly.

The z-score test provides a continuous approximation of the exact binomial test based on the normal distribution. Therefore, it follows the same frequentist approach, assuming that the calibrated PD represents the true PD.

Jeffreys Test

The p-value of the Jeffreys test used to assess evidence of underestimation of the calibrated PD can be calculated as:

$$Pr(X \leq d) = \beta(q = PD, shape1 = nd + 0.5, shape2 = no - nd + 0.5)$$

where:

- β is the cumulative distribution function of the beta distribution;
- PD is the calibrated PD;
- nd is the number of defaults;
- no is the number of observations.

It should be noted that the Jeffreys test follows a Bayesian approach and is therefore naturally formulated as a posterior probability statement. In other words, the Jeffreys test combines the prior information, represented by 0.50 added to both parameters ($shape1$ and $shape2$), with the observed data (nb defaults out of no observations) to obtain the posterior distribution of the true PD. The test then assesses the posterior probability that the true PD is greater than the calibrated PD.

Unlike the frequentist exact binomial and z-score tests, where the calibrated PD is treated as the true PD under the null hypothesis and the observed defaults are evaluated against its sampling distribution, the Jeffreys approach represents the true PD by a posterior distribution derived from the observed data. The calibrated PD is then treated as a reference value against which this posterior distribution of the true PD is evaluated.

Simulation Study

The purpose of this simulation is to visualize the different underlying ways of formulating the null hypothesis in the three most commonly used statistical tests for PD model backtesting.

The simulation is based on the following assumptions for assessing evidence of underestimation of the PD model estimate (calibrated PD):

- the calibrated PD (pd) is set to 1.80%;
- the number of observed defaults (nd) at the testing reference date is 25;
- the number of obligors (no) at the testing reference date is 1,000.

Note:

Given the differences between the presented testing approaches, practitioners are encouraged to further explore the appropriateness of each test under different simulation designs, while also considering the specific nature of IRB model development and calibration. Furthermore, by comparing the Bayesian approach of the Jeffreys test with the frequentist approaches of the exact binomial and z-score tests, practitioners can investigate their respective coverage properties and assess the impact of applying each test at the portfolio or segment level, where PD estimates typically combine heterogeneous estimates across rating grades.

Simulation Results

